

*****Draft*****
Amplitude Shift Keying (ASK) Raspberry Pi
07/22/26
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could not fork this repo <https://github.com/hallard/RadioHead.git>
now included in <https://github.com/develone/raspi-ask-433-tests>

To use the **RadioHead** ASK (Amplitude Shift Keying) library for a 433/315 MHz receiver on a Raspberry Pi in C, you must rely on the wrapper for the underlying C++ library and the bcm2835 hardware driver. However, due to the Pi's multitasking OS, it cannot handle the strict, cycle-accurate bit-banging required for basic ASK, so you must use hardware-accelerated modules like the **RFM69** or **RFM95**

Because the standard ASK classes in the [RadioHead Packet Radio Library](#) were originally designed for microcontrollers with dedicated timers, using cheap ASK receivers (like the FS1000A) natively on the Raspberry Pi often results in poor performance. Instead, connecting an Arduino to your Pi via USB/Serial to do the radio heavy-lifting is the community's preferred workaround

Setting Up the Environment (RFM Modules)

If you are using a supported module (e.g., RFM69) wired to the Pi's SPI pins, you will need to compile the C++ example using the bcm2835 library

```
cd bcm2835-1.75
./configure make
sudo make check
sudo make install
```

Repo <https://github.com/develone/raspi-ask-433-tests>

```
transmitter
sudo apt update
sudo apt-get install pigpio
```

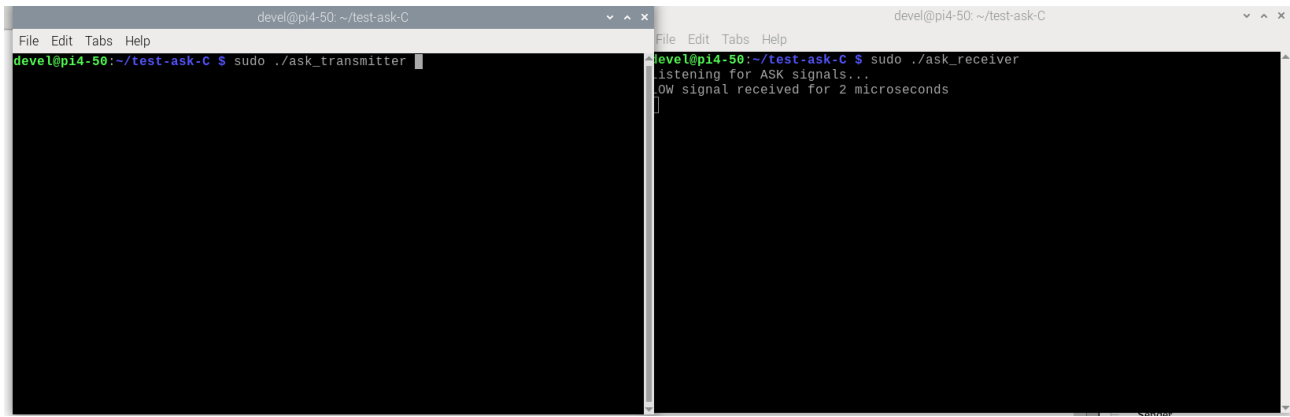
```
gcc -Wall -pthread -o ask_transmitter ask_transmitter.c -lpigpio -lrt
```

```
devel@pi4-50:~/test-ask-C $ sudo ./ask_transmitter
```

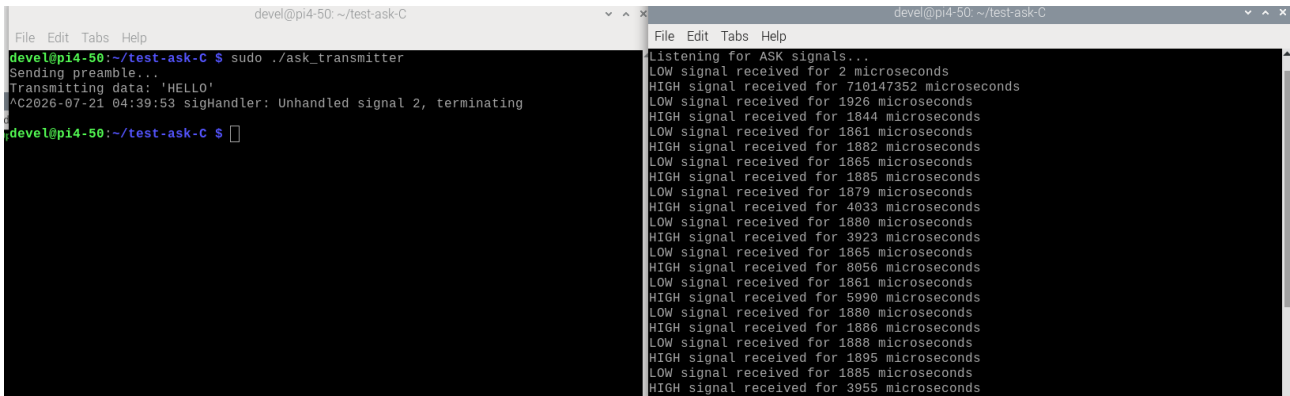
```
receiver
Requires WiringPi https://github.com/WiringPi/WiringPi.git
git clone https://github.com/WiringPi/WiringPi.git
cd WiringPi
./build
```

```
gcc -o ask_receiver ask_receiver.c -lwiringPi
devel@pi4-50:~/test-ask-C $ sudo ./ask_receiver
```

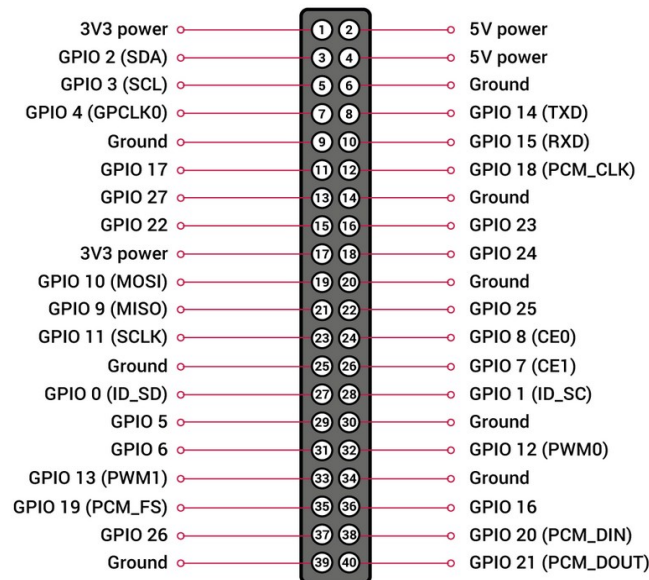
```
transmitter
```



receiver



pi-connector



Needs number of char sent
Needs a CRC